

Filament Routing Attachment

Design and Manufacture Report

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July 27, 2026

Printer	Ender-3 V3 SE
Material	PLA
CAD Tool	FreeCAD
Version	1.0

Part of an open-source engineering portfolio.

GitHub: <https://github.com/Bren10/Filament-Routing-Attachment>

1 Introduction

Problem I use multiple filaments for my Ender-3 V3 SE, often months go by between uses. Specific types of filament, such as PLA, PETG and nylon degrade as the filament absorbs moisture. To remove the moisture I use a Sunlu filament dryer along with my Ender 3. However, the filament heater positioned beside the printer requires routing filament from table height up and back down to the extruder inlet. The stock filament holder on the Ender 3 does not accommodate an elevated feed path, causing the filament to bind at the entry point or collide with larger prints. My current setup can be seen in Fig.1.



Figure 1: My Current Ender-3 V3 SE setup

This report documents the design, manufacture, and testing of a 3D printed filament guide attachment that mounts in place of the stock holder and provides a smooth, low-friction feed path for the elevated filament routing configuration.

2 Design

2.1 Constraints and Goals

The following constraints were identified prior to design:

- Must mount to the existing filament holder location with no permanent modification to the printer frame.
- Must accommodate filament [diameter, 1.75 mm] without excessive clearance that could cause buckling.
- Bend radius must be sufficient to prevent filament binding or fatigue cracking during continuous use.

- Should print without support material where possible to minimise post-processing.
- Must be set high enough so the filament line cannot collide with larger prints.
- Must be able to withstand the partial weight of the dryer, approximately 1 kg (Sometimes filament gets tangled inside the dryer and pulls the dryer over.).
- Must be able to resist heat that is generated from the printing of parts on the Ender 3.
- Must allow for a minimum radius of 35mm. This number was taken from the measurement of a spool.

2.2 Design Concepts

There are two major considerations for the filament design that need to be considered. The interface with the Ender-3 V3 SE, and how the filament is to be guided. These considerations will be evaluated independently then combined together as the final design.

2.2.1 Frame Mounting Design

Twist Lock One concept for the attachment of the Filament Guide would be to reuse the original handle that holds up the filament spool. This could be a simple and proven design that has been shown to work. The original attachment as well as the printed concept can be seen in Fig.2.

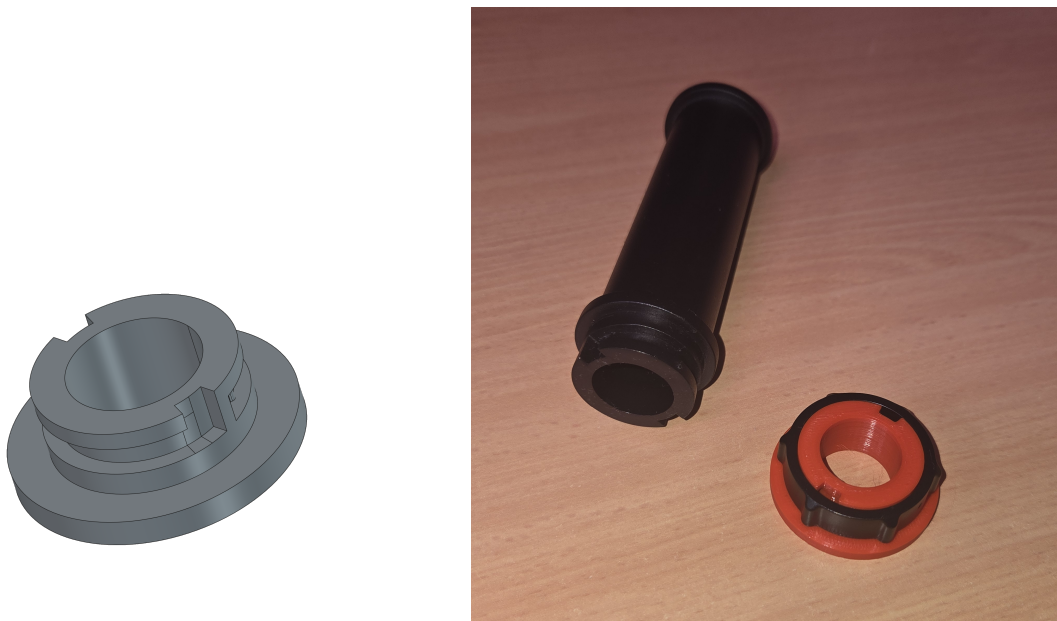
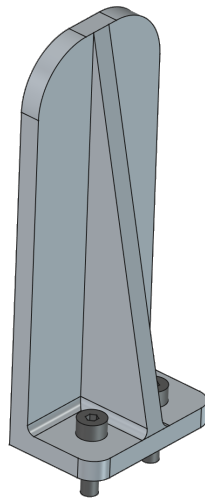


Figure 2: Twist Lock CAD design recreation of the original Ender-3 V3 SE interface

Table 1: Twist Lock Pros and Cons table

Pros	Cons
• Proven Design • Inner hole radius can be decreased to increase strength.	• Requires support • Required accurate tolerance between nut and attachment.

L-Bracket The second concept would be to replace the original L-Bracket on the Ender-3 V3 SE with a 3D printed part. It would reuse the bolt holes of the original bracket. This design would take advantage of the original printer interface and can be removed via unbolting. A representation of the design can be seen in Fig.3

**Figure 3:** L-Bracket Design CAD model**Table 2:** L-Bracket Pros and Cons table

Pros	Cons
• Sturdy • Possible to modify geometry for additional strength • Could place 2D cosmetic design on flat face.	• If printed with the filament mount in one print, would require significant support. • If printed in two parts, would need to consider additional attachment method • Relatively large print if the filament mount is included.

2.2.2 Routing Guide Design

Fixed Pulley Guide The first concept that was considered was to recreate the original spool shape, as can be seen in Fig.4. To reduce filament friction and to make it more

printable, the spool has a round inner surface. I have also considered two designs for the spool, the full rotation and the half rotation. The full rotation will be needed if the attachment is not fixed axially, like the Twist Lock interface. The half rotation can be used if it is, like the L-Bracket.

Table 3: Fixed Pulley Guide Pros and Cons table

Pros	Cons
• Standard Design • Sturdy • Can be used with both Printer interfaces.	• Would require some support to be used. • Large print • Need to consider the friction between the filament and the guider

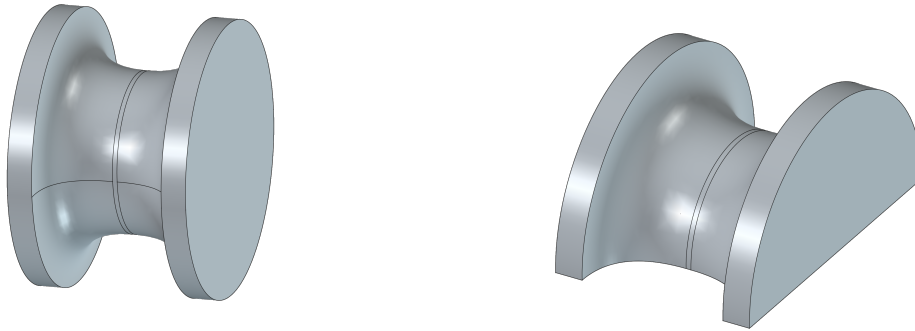
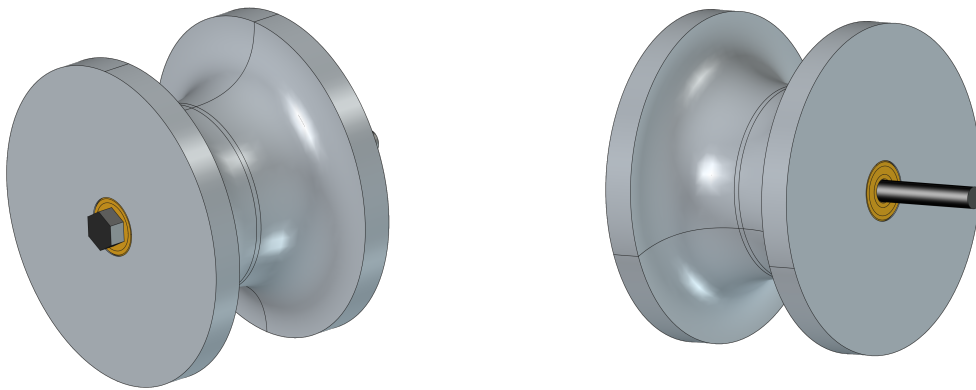


Figure 4: Fixed Pulley Guide CAD Design

Pulley Guide A technique that I have yet to implement in my 3D designs is the use of bearings and I think it could potentially be applicable for this project. An image of the design can be seen in Fig.5. The design would be the same as the full-revolution fixed pulley, however with a bearing in the centre to allow for free rotation. The ball bearing considered would be a standard 608SS bearing. This would be sourced from an old skateboard that has ABEC-7 608ZZ bearings. A hex bolt would go through the centre of the spool and act as an axle, attaching to the two bearings and attaching to the frame mount. The load would be concentrated at the axle-to-frame-mount interface. The frame mount would have to take the cantilever force of the spool. To ensure this does not fail, testing would need to be done.

Table 4: Pulley Guide Pros and Cons table

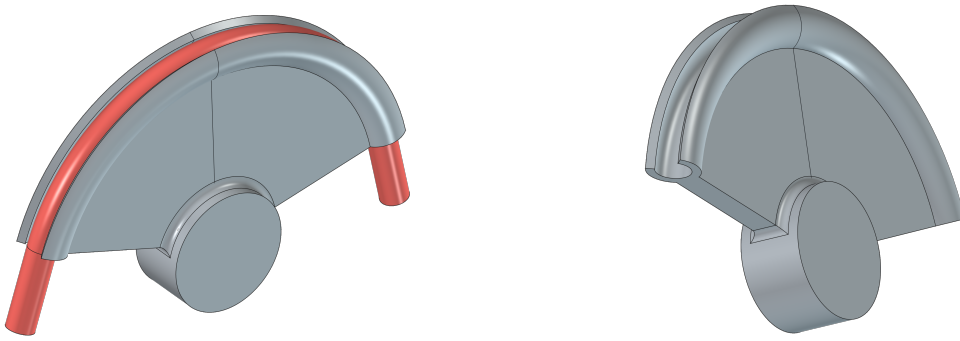
Pros	Cons
• Common Design • Sturdy Spool • Minimal friction on filament	• Would require semi-complex assembly, nut, bolt, bearings etc. • Need to interface the bolt with the Frame mount design. Allowing the axle to be clamped to the Frame mount design while not touching the spool. • Would require a sturdy Frame Mount Design as there will be a strong cantilever force on the bolt.

**Figure 5:** Pulley Guide CAD design

Bracket-Supported Tube A tube could be implemented that could be guided by a 3D printed component. This would allow for a predefined path around the filament holder. A concept of the design can be seen in Fig.6. The red tube in the component would be a Bowden tube that would extend up to the top of the gantry frame. As the extruder would be moving relative to the Filament Guide, the tube cannot be directly attached to the extruder. The tube would need to be securely in place to ensure that the tube does not fall onto the extruder, or cause issues elsewhere.

Table 5: Bracket-Supported Tube Pros and Cons table

Pros	Cons
• Minimal Friction on filament as the tube is designed for filament • Filament cannot slip out of the tube	• Requires installation and secure installation of the tube. • Tube cannot connect to extruder like a real Bowden Tube.

**Figure 6:** Bracket-Supported Tube CAD model

2.2.3 Design Concepts Conclusion

Routing Guide Design Choice The Routing Guide Design was chosen first as this selection directly affects what frame mount is suitable. When considering the Pros and Cons of each Routing Guide, the Fixed Pulley Guide was selected over the regular Pulley Guide and the Bracket-Supported Tube. The Bracket-Supported Tube would be a complex design, requiring the tube to be held securely, if not the printer may be damaged by the tube. Without the tube, the design is essentially the fixed pulley design. As for the fixed vs regular pulley system, the advantage of the bearings would be to reduce the friction on the filament. I do not think the friction will be an issue, if required, the filament may be sanded to reduce friction. Considering these factors, the fixed pulley has been selected.

Frame Mounting Choice When comparing the Twist Lock, to the L-Bracket, the Twist Lock has been shown to be a standard design choice. To change between the Twist Lock, back to the original filament it requires minimal effort. The Routing Guide that was selected does not require a fixed axes so the Twist Lock is acceptable. The L-Bracket and the fixed pulley design would be a very large design if printed together. I would also be concerned with the strength of the L-Bracket, as it would be a tall and narrow. So for these reasons, I have selected the Twist Lock.

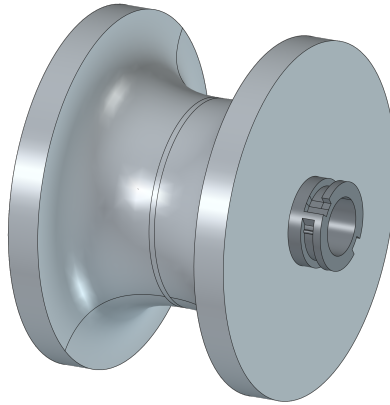


Figure 7: Design concept conclusion selection

Design Concepts Conclusion The final selection that was picked was the Twist Lock interface, and the Fixed Pulley Guide routing guide as can be seen in Fig.7. The design is almost identical to the make-shift version that I currently use as seen in Fig.1. Overall this design is not the most interesting or complex version that I could have designed, however, I do feel as it is the most practical.

2.3 Material Selection

I have not explored the different material that are compatible with my Ender-3 V3 SE. So far I have exclusively used PLA for my designs. I think now would be appropriate to consider PETG as a potential material.

The requirements of the design can be seen in Sec.2.1, the most relevant constrains from the material perspective is the heat resistance, ductility and strength.

Table 6: Material Properties

Property	PLA	PETG
Tensile Strength [1][2]	50-60 MPa	40-50 MPa
Heat Deflection Temperature (HDT) [1]	55°	70°
Chemical / UV Resistance [1]	Low	Moderate
Ductility (% Elongation)[2]	6%	20%
Impact resistance[1]	Lower	Moderate
Printability[1]	High	Moderate

When comparing the properties of the material for the design, the PETG and PLA both have advantages and disadvantages. The key material properties for the design are the heat resistance, ductility and strength. The properties from Tab.6 will be compared. It should be noted that these values are just common ranges and depend on the specific filament.

- **Ductility** - Should the design fail, having a ductile material is preferable as it shows visible signs of failure before catastrophic failure. PETG is significantly more ductile than PLA[2]. This is key as if failure occurs it would fall damaging both the current print and the Ender 3 printer itself.
- **Tensile strength** - PLA and PETG both have comparable tensile strengths. PLA has a slightly higher tensile strength on average[1], however in some cases can be comparable[2].
- **Heat Deflection Temperature** - The heat deflection temperature (HDT) is important due to the proximity of the filament holder to the extruder. PETG has higher HDT than PLA. PLA and PETG have a HDT of 55°, and 70° respectively [1]. It is unlikely that the filament holder would experience temperatures this high as the nozzle would be continuously moving, and on the other side of the extruding head.
- **Impact Resistance** - During printing, sometimes the filament gets tangled and causes a knot, however, the printer still continues to extrude the filament. As the filament dryer is pulled, it sometimes falls, causing pulling on the filament guide. While this is not common, and the filament roll and the dryer are not heavy (2 kg), this is still a possibility that could induce catastrophic failure. PETG is more impact resistant than PLA [1].

When considering the material properties of PLA and PETG, PETG is more suited for this application. The main driving factor is the ductility of PETG. This allows for visible signs of failure before catastrophic failure. It has comparable strength to PLA, and superior HDT. Additionally it also has better impact resistance which might be applicable if the filament gets knotted.

3 Manufacturing

3.1 Print Settings

3.2 Post-Processing

[Describe any post-processing here, or note that none was required.]

4 Results

4.1 Version 1

[Describe v1 results here.]

4.2 Version 2 (if applicable)

[Describe v2 changes here, or remove this subsection if v1 was final.]

5 Conclusions

[Summarise whether the guide solved the original problem, any limitations of the current design, and potential future improvements.]

6 Future Improvements

[What should I improve upon for next time]

- Tolerances of print (Lost on Windows version)
- Remember what you were going to write and put it here

References

- [1] *PETG vs PLA vs ABS: 3D Printing Strength Comparison*, <https://ultimaker.com/learn/petg-vs-pla-vs-abs-3d-printing-strength-comparison/>. (visited on 07/24/2026).
- [2] *What is PLA filament Simple Guide to Key Properties*, <https://www.sovol3d.com/blogs/news/what-is-pla-filament-key-properties-guide-3d-printing>. (visited on 07/25/2026).